

Kafaah Industrial Solutions — Insights

01 — STRATEGIC

Why the Most Expensive Mistakes in Chemical Plants Happen After Construction Ends

Most project budgets are built around engineering and construction. The assumption is simple: once the plant is built, the hard part is over.

It isn't.

The commissioning and startup phase — the period between mechanical completion and stable, on-spec production — is where the majority of lifetime performance gaps are created. Not because the engineering was wrong. Not because the equipment failed. But because the transition from a constructed asset to an operating plant is a discipline in itself, and it is almost never treated that way.

Plants that commission without process-specific operational expertise routinely experience yield gaps in their first year of operation. Tuning cycles that should take weeks stretch into months. Equipment that was installed correctly fails to perform at design intent because the operating envelope was never properly established during startup.

The cost compounds. Every month of suboptimal operation is not just lost revenue — it is a baseline that the plant may never recover from. Operators learn to work around problems rather than solve them. Workarounds become procedures. Procedures become the new normal.

The solution is not more engineering. It is operational expertise applied at the right moment — during commissioning, during startup, and during the critical first months of production. Engineers who have operated plants like yours, not engineers who have designed them.

This is the phase Kafaah focuses on. Not because it is the most visible — but because it is where the difference between a plant that performs and a plant that underperforms is made.

02 — TECHNICAL

What Commissioning Actually Means — And Why Most Projects Get It Wrong

Commissioning is one of the most misused terms in industrial project management. Ask ten project managers to define it and you will get ten different answers — most of them wrong in ways that cost money.

In practice, commissioning is often treated as a final checklist: verify that equipment is installed, run a pressure test, flush the lines, and declare readiness for startup. This approach mistakes

mechanical completeness for operational readiness. They are not the same thing.

True commissioning is a systematic process of transforming individual installed systems into an integrated, operating plant. It requires understanding not just whether each component works in isolation — but how the plant behaves as a system under real process conditions.

This distinction matters most in inorganic chemical and fertilizer plants, where process interactions are complex and the margin for error during startup is narrow. A sulfuric acid plant that has been mechanically completed but not properly pre-commissioned will expose its gaps the moment sulfur combustion begins. A phosphoric acid plant with inadequately verified instrumentation will struggle to maintain the process parameters that determine product quality.

The failures are rarely dramatic. They appear as instability — temperatures that drift, concentrations that won't hold, equipment that cycles when it should run steadily. Each symptom has a root cause. But finding it during startup, under pressure to produce, is expensive.

Proper commissioning prevents this by doing the diagnostic work before it matters. It is not a checklist. It is a methodical process of building confidence — system by system, loop by loop — that the plant is ready to operate.

03 — STRATEGIC

What an Owner's Engineer Actually Does — And What Happens Without One

When a project owner contracts an EPC company, they are buying a promise: deliver a working plant, on time, within budget, to specification. The EPC's job is to fulfill that promise as efficiently as possible — which means managing cost, schedule, and constructability from their perspective.

That perspective is not the same as the owner's.

An EPC company manages dozens of projects simultaneously. They have established relationships with equipment vendors, preferred engineering approaches, and contractual incentives that may not always align with long-term plant performance. This is not a criticism — it is simply the nature of how EPC contracts work. The EPC is optimizing for project delivery. The owner needs to optimize for a plant that performs reliably for the next twenty years.

Without independent technical representation, the owner's interests in that equation are undefended.

An Owner's Engineer fills that gap. Not as an adversary to the EPC — but as the owner's technical voice throughout the project. Reviewing engineering deliverables. Questioning assumptions that could affect operability. Monitoring construction quality. Ensuring that the

commissioning and startup phase receives the attention it deserves rather than being compressed to meet handover deadlines.

The value is difficult to quantify in advance and obvious in retrospect. Owners who have experienced a poorly executed handover — a plant that technically met contract specifications but failed to perform at design intent — understand exactly what independent technical oversight is worth.

Those who haven't yet had that experience often question whether they need it. The answer is that they do — they simply haven't paid for the lesson yet.

04 — TECHNICAL

The Sulfuric Acid Plant Startup Sequence: Where Things Go Wrong and Why

A sulfuric acid plant using the contact process is, in principle, well-understood technology. The chemistry is established. The equipment configurations are mature. Licensors provide detailed operating manuals.

And yet, sulfuric acid plant startups — particularly in facilities with less experienced operations teams — are among the most operationally demanding in the inorganic chemicals sector.

The reason is not the chemistry. It is the sequence.

A contact process H_2SO_4 plant must be brought to operating conditions in a carefully controlled order. The converter must reach light-off temperature before sulfur combustion begins. The catalyst beds must be protected from condensation. The absorption system must be ready to handle SO_3 before it is produced. Each step depends on the previous one being correctly executed.

Deviation from the sequence — rushing a step, misjudging a temperature, starting flows in the wrong order — does not always produce an immediate visible failure. Instead, it creates conditions that degrade performance gradually. Catalyst poisoning that reduces conversion efficiency over time. Absorber upsets that affect acid concentration. Emission control systems that struggle to maintain compliance under off-design conditions.

The most common startup errors we encounter are not the dramatic ones. They are the subtle ones: insufficient pre-commissioning of instrumentation loops, inadequate purging before light-off, temperature ramp rates that are too aggressive, and operators who have been trained on procedures rather than on process understanding.

Process understanding is the difference. An operator who knows why the sequence matters — not just what the sequence is — can respond correctly when conditions deviate from the manual. That knowledge comes from experience inside operating plants, not from classroom training alone.

05 — STRATEGIC

The Real Cost of Delayed Ramp-Up: A Framework for Project Owners

Every project financial model includes a production ramp-up curve. Most assume that the plant will reach design capacity within a defined period — typically three to six months after first production.

Most are wrong. And the financial consequences of being wrong are rarely visible until they have already occurred.

A delayed ramp-up does not appear as a single large cost. It appears as a series of smaller, ongoing losses that accumulate over months: lower yields than modeled, higher energy consumption per unit of product, increased maintenance interventions, and operator overtime that was not in the budget. Each item is manageable in isolation. Together, they reshape the project's return profile in ways that are difficult to recover from.

The root cause is almost always the same: the transition from construction to operation was not treated as a technical discipline. It was treated as a schedule milestone.

The framework for avoiding this is straightforward:

First, commission the plant properly — not just mechanically, but operationally. Verify that every system behaves as designed before startup begins.

Second, staff the startup with engineers who have operated this specific type of plant before. The learning curve of a team encountering a sulfuric acid or phosphoric acid plant for the first time during startup is expensive. Someone who has done it before has already paid that cost on another project.

Third, establish clear performance baselines early. Know what the plant should be producing, at what efficiency, under what operating conditions — and track against those baselines from day one.

Fourth, plan for problems. They will occur. The difference between a ramp-up that recovers quickly and one that drags for months is the speed and quality of the diagnostic response when something deviates.

These are not complex principles. They are consistently underimplemented because the project organization that built the plant is not the same organization that will operate it — and the handover between the two is rarely given the attention it deserves.

06 — TECHNICAL

Understanding the Mannheim Process: Operational Challenges and Startup Considerations

Potassium sulfate production via the Mannheim process is among the more operationally demanding fertilizer technologies in widespread commercial use. The process is well-established — but its steady-state operation requires a level of process discipline that is frequently underestimated during project planning.

The Mannheim furnace operates at high temperatures with a corrosive process environment. The reaction between potassium chloride and sulfuric acid produces potassium sulfate and hydrochloric acid gas simultaneously, requiring integrated management of both product streams from the moment the process begins.

Startup presents specific challenges that differ from steady-state operation:

Furnace heat-up must be managed carefully to avoid thermal shock to refractory materials and to establish the correct temperature profile across the furnace bed before reagent introduction begins. Rushing this phase is one of the most common sources of early operational problems.

Reagent introduction must be staged to allow the process to stabilize incrementally. The interaction between feed rates, furnace temperature, and product quality is not linear — small deviations early in startup can produce significant product specification problems that take time to correct.

HCl absorption system readiness is critical and frequently inadequately verified during pre-commissioning. The absorption system must be capable of handling the HCl load from the moment reagent introduction begins. Gaps in absorption efficiency during startup create both product quality and environmental compliance problems.

Product handling — the miller and conveying system — must be integrated with furnace operations from the beginning. Blockages or handling failures during startup create back-pressure on the process that can disrupt the entire operation.

The SOP plant we commissioned in Suez illustrated all of these challenges in sequence. The methodical pre-commissioning of each system — verified before the sequence began rather than discovered during it — was the difference between a stable startup and a prolonged stabilization period.

07 — STRATEGIC

How EPC Contractors Can Protect Themselves During Commissioning and Startup

EPC contractors carry a concentration of risk during commissioning and startup that is disproportionate to the value they receive from it. The contract has been largely executed. The

engineering is done. The construction is complete. But contractual obligations — performance guarantees, handover milestones, defect liability — remain fully active during a phase that the EPC organization is often ill-equipped to manage.

The core problem is structural. EPC companies are built for engineering and construction. Their project organizations are staffed accordingly. The process specialists, commissioning engineers, and operational experts needed to execute startup effectively are not typically part of the permanent EPC workforce — they are mobilized for specific projects, often at short notice, often without deep familiarity with the specific plant technology involved.

This creates exposure. A startup that drags extends site costs. Performance tests that fail delay final payment. Equipment that fails to reach design intent generates defect claims. Each of these outcomes is both financially material and reputationally damaging.

The mitigation is to treat commissioning and startup as a discipline that requires specific expertise — not as the final phase of construction. This means:

Engaging process specialists who have operated the specific plant technology before, not generalists who will learn on the project. Establishing a commissioning organization that is separate from and supported by the construction organization, with clear handover protocols between them. Pre-commissioning every system to a defined standard before startup activities begin. Having an operational troubleshooting resource available and ready to respond when — not if — startup deviations occur.

The cost of this expertise is a fraction of the cost of a startup that goes wrong. EPC contractors who have experienced a difficult handover understand this. Those who haven't yet are managing risk they may not have fully priced.

08 — TECHNICAL

NPK Fertilizer Production: Why Granulation Lines Fail to Reach Design Capacity

NPK fertilizer granulation is, on paper, a relatively forgiving process. The technology is mature. The equipment is well-understood. The operating parameters are less extreme than those found in acid plants or potassium sulfate production.

In practice, NPK granulation lines — particularly drum granulation and compaction granulation systems — consistently fail to reach design capacity in the first months of operation, and the reasons are almost always the same.

Moisture management is the most common source of instability. Granulation is sensitive to small variations in feed moisture, ambient conditions, and recycle ratios. A line that is not properly tuned for its specific raw material characteristics and operating environment will produce off-spec product that increases recycle, reduces throughput, and degrades overall

efficiency. Getting this balance right requires operational experience — not just process design knowledge.

Recycle ratio control is closely related. Most granulation lines are designed to operate within a specific recycle range. Lines that startup with inadequately defined control strategies struggle to find and maintain the operating point that produces acceptable product at design throughput.

Screen performance is frequently underestimated during commissioning. Screen blinding, wear, and inappropriate aperture selection for the specific product being produced are common sources of throughput loss that are easy to correct — but only if they are correctly diagnosed.

Dryer operation interacts with granulation in ways that are not always intuitive. Dryer exit moisture targets that seem conservative on paper may prove inadequate for the specific product bulk density and storage conditions involved.

The compaction granulation line we supported in Yanbu encountered several of these challenges during startup. The diagnostic approach — systematic rather than reactive, focused on understanding the process behavior rather than chasing individual symptoms — allowed stable operation across multiple fertilizer grades to be established within the planned timeline.

09 — STRATEGIC

The Handover Problem: Why Plants Underperform After EPC Completion

There is a structural discontinuity at the heart of most industrial plant projects. The organization that builds the plant and the organization that will operate it are different — and the transfer of knowledge, responsibility, and operational understanding between them is consistently undermanaged.

This is the handover problem. And it is responsible for more long-term plant underperformance than any other single factor.

The symptoms are familiar: a plant that technically passes its performance tests but struggles to maintain design output under normal operating conditions. An operations team that knows how to follow procedures but lacks the process understanding to respond effectively when conditions deviate. A maintenance organization that is reactive rather than predictive because the operational baseline was never properly established.

The root cause is that handover is treated as a legal and contractual event rather than a knowledge transfer event. Documents are signed. Certificates are issued. The EPC demobilizes. The operations team takes over. And the gap between what the plant can do in theory and what it actually produces in practice slowly becomes the new normal.

Closing this gap requires deliberate investment in three areas:

Operational knowledge transfer — not just training on procedures, but systematic development of process understanding within the operations team. Operators who understand why the plant behaves as it does can troubleshoot. Operators who only know what to do in normal conditions cannot.

Baseline establishment — defining and documenting what the plant actually produces under normal operating conditions during the first weeks of operation, before problems accumulate and before the operations team adapts to workarounds.

Early support — having process expertise available and engaged during the first months of operation, when deviations from design are most common and most consequential.

None of this is complex in principle. It is consistently deprioritized in practice because the project organization that should be delivering it is already moving to the next project.

10 — TECHNICAL & STRATEGIC

What Twenty Years Inside Chemical Plants Teaches You That Engineering Textbooks Don't

Process engineering education is excellent at teaching the theory of chemical plant operation. Thermodynamics. Reaction kinetics. Mass and energy balances. Separation processes. Control theory. A well-trained process engineer graduating today has access to better analytical tools and more comprehensive technical literature than any generation before them.

What the education cannot provide is the knowledge that comes from standing inside an operating plant when something goes wrong — and finding the answer.

This is not a criticism of engineering education. It is a description of a gap that only operational experience can fill.

After nearly two decades inside sulfuric acid, phosphoric acid, potassium sulfate, NPK, magnesium sulfate, and SSP plants — across commissioning, startup, troubleshooting, and optimization — the lessons that matter most are not the ones in the textbooks.

Plants behave as systems, not as unit operations. A problem in the absorber of a sulfuric acid plant may have its root cause in the converter. An NPK granulation instability may originate in feed preparation. Isolating symptoms and fixing them in isolation misses the process interaction that caused them.

Operators know things that instruments don't measure. Experienced operators develop sensitivity to plant behavior — sounds, smells, subtle visual cues — that no instrumentation captures. The best troubleshooting integrates data with operational observation.

The design is a starting point, not an answer. Every plant operates differently from its design basis. Feedstock variations, equipment wear, seasonal conditions, and operational history all

push the real operating envelope away from the theoretical one. Optimization requires understanding where the real envelope is, not where the design said it would be.

Speed of diagnosis matters more than comprehensiveness. In a plant that is not performing, every day of suboptimal operation has a cost. A good diagnosis that arrives quickly and is 80% correct is worth more than a comprehensive analysis that takes three times as long.

Trust is built by being right. Operations teams learn quickly whether outside expertise adds value or adds noise. Credibility comes from making correct calls — not from credentials or reports.

These are the things that twenty years teaches. They cannot be shortcut. But they can be brought to your project — which is why Kafaah exists.

Kafaah Industrial Solutions — Insights reflect the direct operational experience of our founder and specialist network across inorganic chemical and fertilizer plant commissioning, startup, troubleshooting, and optimization.